

Project Summary

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

Overview

TriadicFrameworks is an open-science research initiative developing a unified substrate for regime-aware scientific analysis. The project provides openly licensed protocols, models, and tools that enable consistent interpretation of structural, sensory, and environmental patterns across sensing systems, simulations, and scientific workflows. This proposal seeks HPOSS support to expand TriadicFrameworks into a fully documented, community-accessible infrastructure that advances NASA's mission through transparent, reproducible, and interoperable open-source science.

Scientific and Technical Objectives

The proposed work focuses on four core components:

- 1. Dimensional Substrate Regime Scanning Protocol (dsrsp/0.1)**

A minimal, substrate-agnostic wire format for regime-aware sensor interpretation. Enables consistent classification of structural, sensory, and environmental regimes across diverse platforms, including remote sensing, autonomous systems, and simulation environments.

- 2. Resonance Substrate Model (RSM)**

A structural grammar for representing coherence, drift, and stability across physical and simulated systems.

Provides a foundation for reproducible structural analysis and cross-domain comparison.

3. Validation-Space-Time Engine (vST)

A validation framework for analyzing regime transitions, dimensional continuity, and alignment behavior.

Supports both real-world sensing pipelines and simulation-based training environments.

4. Structural Life-Regime Profiles (SLRP)

A standardized set of life-regime exemplars for biological, synthetic, and planetary systems.

Enables consistent interpretation of life-like signatures and supports optional intelligent-life probability (ILP) analysis.

Together, these components form a coherent open-science substrate that supports NASA's goals in remote sensing, planetary science, autonomous exploration, and scientific reproducibility.

Relevance to NASA's Open-Science Mission

TriadicFrameworks directly advances HPOSS priorities by:

- providing openly licensed scientific infrastructure,
- enabling reproducible workflows for sensor interpretation and regime classification,
- supporting transparent, community-accessible scientific tools,
- improving interoperability across missions and research domains, and
- lowering barriers for researchers, educators, and developers to engage in open science.

The framework is designed to integrate with NASA's existing open-science ecosystem, including data repositories, simulation environments, and remote-sensing workflows.

Work Plan and Deliverables

Over the 12-month performance period, the project will deliver:

- fully documented dsrsp/0.1, RSM, vST, and SLRP specifications,
- open-source reference implementations and libraries,
- example datasets and reproducible workflows,
- developer-ready schemas and integration guides,
- community documentation and onboarding materials, and
- a public release of the TriadicFrameworks open-science substrate.

All deliverables will be released under permissive open licenses and archived with DOIs to ensure long-term accessibility.

Expected Impact

TriadicFrameworks will provide NASA and the broader scientific community with a robust, extensible foundation for regime-aware analysis across sensing, modeling, and simulation. By establishing a shared substrate for structural interpretation and validation, the project enhances scientific transparency, accelerates tool development, and supports the long-term goals of NASA's Open-Source Science Initiative.

This proposal positions TriadicFrameworks as a scalable, community-driven infrastructure capable of supporting future NASA missions, open-science collaborations, and cross-disciplinary research.